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Semiconductor package including plurality of semiconductor chips and method for manufacturing the same

Abstract

A semiconductor package is disclosed. The semiconductor package includes a base structure, a first semiconductor chip over the base structure, a second semiconductor chip over the first semiconductor chip, an adhesive layer between the first semiconductor chip and the second semiconductor chip, and a molding layer covering the first semiconductor chip, the second semiconductor chip and the adhesive layer, and including an interposition portion interposed between the base structure and the first semiconductor chip.

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/384,046, filed on Jul. 23, 2021, now U.S. Pat. No. 11,935,868, which claims priority from Korean Patent Application No. 10-2021-0003240, filed on Jan. 11, 2021, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(1) The example embodiments of the disclosure relate to a semiconductor package and a method for manufacturing the same. More particularly, the example embodiments of the disclosure relate to a semiconductor package including a plurality of semiconductor chips and a method for manufacturing the same.

2. Description of the Related Art

(2) In accordance with advances in electronics and demand of users, electronic appliances become further miniaturized and lightened. Semiconductor packages used in such electronic appliances are required to have high performance and a large capacity in addition to miniaturization and lightness. In order to achieve high performance and a large capacity as well as miniaturization and lightness, research and development of semiconductor chips including a through-silicon via (TSV) and a semiconductor package, in which the semiconductor chips are stacked, is being continuously conducted.

SUMMARY

(3) The example embodiments of the disclosure provide a semiconductor package having enhanced reliability.

(4) A semiconductor package according to some example embodiments of the disclosure may include a base structure, a first semiconductor chip over the base structure, a second semiconductor chip over the first semiconductor chip, an adhesive layer between the first semiconductor chip and the second semiconductor chip, and a molding layer covering the first semiconductor chip, the second semiconductor chip and the adhesive layer and including an interposition portion interposed between the base structure and the first semiconductor chip.

(5) A semiconductor package according to some example embodiments of the disclosure may include a base structure, a first semiconductor chip over the base structure, a first connecting structure electrically interconnecting the base structure and the first semiconductor chip, a second semiconductor chip over the first semiconductor chip, a second connecting structure electrically interconnecting the first semiconductor chip and the second semiconductor chip, an adhesive layer between the first semiconductor chip and the second semiconductor chip, and a molding layer covering the first semiconductor chip, the second semiconductor chip, and the adhesive layer, and including an interposition portion contacting the first connecting structure.

(6) A semiconductor package according to some example embodiments of the disclosure may include a base structure, a first semiconductor chip over the base structure, second semiconductor chips over the first semiconductor chip, connecting structures electrically interconnecting the base structure, the first semiconductor chip and the second semiconductor chips, adhesive layers overlapping with the base structure, the first semiconductor chip and the second semiconductor chips, and a molding layer covering the first semiconductor chip, the second semiconductor chips and the adhesive layers. The molding layer may include an interposition portion overlapping with the base structure, the first semiconductor chip, the second semiconductor chips, and the adhesive layers. The interposition portion may contact at least one of the connecting structures and a bottom surface of at least one of the first and second semiconductor chips.

(7) A method for manufacturing a semiconductor package in accordance with some example embodiments of the disclosure may include forming a base structure, providing a first semiconductor chip over the base structure, a cavity being defined between the base structure and

the first semiconductor chip, providing a second semiconductor chip over the first semiconductor chip using an adhesive layer, and forming a molding layer covering the first semiconductor chip, the second semiconductor chip and the adhesive layer. The forming the molding layer may include filling the cavity with an interposition portion of the molding layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (2) FIG. 1B is an enlarged view of a portion A in FIG. 1A.
- (3) FIG. 1C is an enlarged view of a portion B in FIG. 1B.
- (4) FIGS. 2a, 2b, 2c, 2d, 2e, 2f, 2g, 2h, 2i, 2j, 2k, and 2l are sectional views explaining a method for manufacturing a semiconductor package in accordance with some example embodiments of the disclosure.
- (5) FIG. 3 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (6) FIG. 4 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (7) FIG. 5 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (8) FIG. 6 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (9) FIG. 7 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (10) FIG. 8 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.
- (11) FIG. 9 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (12) One or more example embodiments will be described in detail with reference to the accompanying drawings. Example embodiments, however, may be embodied in various different forms, and should not be construed as being limited to only the illustrated embodiments. Unless otherwise noted, like reference, characters denote like elements throughout the attached drawings and written description, and thus descriptions will not be repeated.
- (13) Although the terms “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections, should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section, from another region, layer, or section. Thus, a first element, component, region, layer, or section, discussed below may be termed a second element, component, region, layer, or section, without departing from the scope of this disclosure.
- (14) Spatially relative terms, such as “below,” “above,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, the device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.
- (15) When the terms “about” or “substantially” are used in this specification in connection with a

numerical value, it is intended that the associated numerical value includes a manufacturing tolerance (e.g., $\pm 10\%$) around the stated numerical value. Moreover, when the words “similar” and “substantially” are used in connection with composition and/or geometric shapes, it is intended that precision of composition and/or the geometric shape is not required but that latitude for the composition and/or shape is within the scope of the disclosure. It will be understood that these values, compositions, and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values, compositions, or shapes.

(16) FIG. 1A is a sectional view of a semiconductor package according to some example embodiments of the disclosure. FIG. 1B is an enlarged view of a portion A in FIG. 1A. FIG. 1C is an enlarged view of a portion B in FIG. 1B.

(17) Referring to FIG. 1A, a semiconductor package **1** may include a base structure **100**. The base structure **100** may include a first substrate **110**, a first wiring structure **120**, a first lower protective layer **130**, a first upper protective layer **140**, and first through vias **150**.

(18) The first substrate **110** may have the form of a plate extending along a plane defined by a first direction **D1** and a second direction **D2**. The first direction **D1** and the second direction **D2** may intersect each other. For example, the first direction **D1** and the second direction **D2** may perpendicularly intersect each other. In some embodiments, the first substrate **110** may be a semiconductor substrate. For example, the first substrate **110** may include at least one of an elemental semiconductor (e.g., silicon, germanium) and/or a compound semiconductor (e.g., silicon-germanium, and/or a III-V semiconductor (e.g., GaP, and/or GaAs)). In some embodiments, the first substrate **110** may be a silicon-on-insulator (SOI) substrate and/or a germanium-on-insulator (GOI) substrate.

(19) The first substrate **110** may be provided on the first wiring layer **120**. The first wiring layer **120** may be provided under the first substrate **110**. The first wiring layer **120** may include wirings, and an insulating layer surrounding the wirings. The wirings of the first wiring layer **120** may include a conductive material.

(20) In some embodiments, the base structure **100** may include a logic circuit, and/or the base structure **100** may include a memory device.

(21) The first wiring layer **120** may be provided on the first lower protective layer **130**. The first lower protective layer **130** may be provided under the first wiring layer **120**. The first lower protective layer **130** may function to protect the first wiring layer **120**. The first lower protective layer **130** may cover a bottom surface of the first wiring layer **120**. The first lower protective layer **130** may include an insulating material. In some embodiments, the first lower protective layer **130** and the insulating layer of the first wiring layer **120** may include the same and/or a different insulating material. In some embodiments, the first lower protective layer **130** may include silicon nitride and/or silicon oxide.

(22) The first upper protective layer **140** may be provided on the first substrate **110**. The first upper protective layer **140** may function to protect the first substrate **110**. The first upper protective layer **140** may cover a top surface of the first substrate **110**. The first upper protective layer **140** may include an insulating material. In some embodiments, the first lower protective layer **130** and the first upper protection layer **140** may include the same and/or a different insulating material. For example, the first upper protective layer **140** may include silicon nitride and/or silicon oxide.

(23) The first through vias **150** may extend through the first substrate **110** and the first upper protective layer **140**. The first through vias **150** may be electrically connected to the wirings of the first wiring layer **120**. The first through vias **150** may include a conductive material.

(24) Although the base structure **100** is shown and described as including the first substrate **110**, the first wiring layer **120**, and the first through vias **150**, the example embodiments of the disclosure are not limited thereto. In some embodiments, the base structure **100** may be and/or include an interposer. The interposer may be configured, for example, to expand the pitch between contacts associated with a first through via **150** on a surface (e.g., a lower surface) of the interposer

compared to the equivalent contacts on another surface (e.g., an upper surface).

(25) Base pads BPD and base bumps BBP may be provided under the base structure **100**. The base pads BPD may extend through the first lower protective layer **130**. The base pads BPD may be electrically connected to the wirings of the first wiring layer **120**. The base pads BPD may include a conductive material. Though illustrated as extending past the first lower protective layer **130**, the example embodiments of the disclosure are not limited thereto, and a bottom surface of the base pads BPD may be level or at a different level with a bottom surface of the first lower protective layer **130**. The base bumps BBP may be connected to the base pads BPD. The base bumps BBP may include a conductive material. For example, the base bumps BBP may include a solder ball and/or pillar, and may include a solder, though the example embodiments of the disclosure are not limited thereto.

(26) A first semiconductor chip **200** may be provided over the base structure **100**. For example, the first semiconductor chip **200** may be provided over the base structure **100** in a vertical direction. The first semiconductor chip **200** may be spaced apart from the base structure **100** in a third direction **D3**. The first semiconductor chip **200** may be a semiconductor chip adjacent to the base structure **100**. The third direction **D3** may intersect with the first direction **D1** and the second direction **D2**. The third direction **D3** may perpendicularly intersect with the first direction **D1** and the second direction **D2** (e.g., the third direction **D3** may be perpendicular to the plane defined by the first direction **D1** and the second direction **D2** and/or the vertical direction).

(27) The first semiconductor chip **200** may include a second substrate **210**, a second wiring layer **220**, a second lower protective layer **230**, a second upper protective layer **240**, and second through vias **250**.

(28) The second substrate **210** may have the form of a plate extending along a plane defined by the first direction **D1** and the second direction **D2**. In some embodiments, the second substrate **210** may be a semiconductor substrate. For example, the second substrate **210** may include at least one of an elemental semiconductor (e.g., silicon, germanium) and/or a compound semiconductor (e.g., silicon-germanium, and/or a III-V semiconductor (e.g., GaP, and/or GaAs)). In some embodiments, the second substrate **210** may be a silicon-on-insulator (SOI) substrate and/or a germanium-on-insulator (GOI) substrate.

(29) The second substrate **210** may be provided on the second wiring layer **220**. The second wiring layer **220** may be provided under the second substrate **210**. The second wiring layer **220** may include wirings, and an insulating layer surrounding the wirings. The wirings of the second wiring layer **220** may include a conductive material. In some embodiments, the first wiring layer **120** and the second wiring layer **220** may include the same and/or different materials.

(30) In some embodiments, the first semiconductor chip **200** may include a memory device. For example, the memory device may be volatile memory such as dynamic random access memory (DRAM) and/or non-volatile memory such as flash memory, magnetic RAM (MRAM), ferroelectric RAM (FRAM), phase change RAM (PRAM), resistive RAM (RRAM), etc. In some embodiments, the first semiconductor chip **200** may include a logic circuit.

(31) The second wiring layer **220** may be provided on the second lower protective layer **230**. The second lower protective layer **230** may be provided under the second wiring layer **220**. The second lower protective layer **230** may function to protect the second wiring layer **220**. The second lower protective layer **230** may face the base structure **100**. The second lower protective layer **230** may face the first upper protective layer **140** of the base structure **100**. The second lower protective layer **230** may cover a bottom surface of the second wiring layer **220**. The second lower protective layer **230** may include an insulating material. The first lower protective layer **130** and the second lower protective layer **230** may include the same and/or a different insulating material. For example, the second lower protective layer **230** may include silicon nitride and/or silicon oxide.

(32) The second upper protective layer **240** may be provided on the second substrate **210**. The second upper protective layer **240** may function to protect the second substrate **210**. The second

upper protective layer **240** may cover a top surface of the second substrate **210**. The second upper protective layer **240** may include an insulating material. The first upper protective layer **140** and the second upper protection layer **240** may include the same and/or a different insulating material. For example, the second upper protective layer **240** may include silicon nitride and/or silicon oxide.

(33) The second through vias **250** may extend through the second substrate **210** and the second upper protective layer **240**. The second through vias **250** may be electrically connected to the wirings of the second wiring layer **220**. The second through vias **250** may include a conductive material.

(34) First connecting structures **CS1** may be provided between the base structure **100** and the first semiconductor chip **200**. The first connecting structures **CS1** may electrically interconnect the base structure **100** and the first semiconductor chip **200**. The first connecting structures **CS1** may include a conductive material. Each of the first connecting structures **CS1** may include an upper pad **UPD**, a lower pad **LPD**, and a connecting bump **CBP**.

(35) The upper pads **UPD** of the first connecting structures **CS1** may be provided on and/or extend through the second lower protective layer **230** of the first semiconductor chip **200**. The upper pads **UPD** of the first connecting structures **CS1** may be electrically connected to the wirings of the second wiring layer **220** of the first semiconductor chip **200**. The upper pads **UPD** of the first connecting structures **CS1** may include a conductive material.

(36) The lower pads **LPD** of the first connecting structures **CS1** may be provided on and/or extend through the first upper protective layer **140** of the base structure **100**. The lower pads **LPD** of the first connecting structures **CS1** may be electrically connected to the first through vias **150** of the base structure **100**. The lower pads **LPD** of the first connecting structures **CS1** may include a conductive material.

(37) The connecting bump **CBP** of each first connecting structure **CS1** may be provided between the upper pad **UPD** and the lower pad **LPD** in the first connecting structure **CS1**. The connecting bump **CBP** of the first connecting structure **CS1** may electrically interconnect the upper pad **UPD** and the lower pad **LPD** in the first connecting structure **CS1**. The connecting bump **CBP** of the first connecting structure **CS1** may include a conductive material. For example, the connecting bumps **CBP** may include a solder ball and/or pillar, and may include a solder, though the example embodiments of the disclosure are not limited thereto.

(38) A plurality of second semiconductor chips **300** may be provided over the first semiconductor chip **200**. For example, the plurality of second semiconductor chips **300** may be provided over the first semiconductor chip **200** in a vertical direction. For example, the plurality of second semiconductor chips **300** may be stacked over the first semiconductor chip **200** in the third direction **D3**. The first semiconductor chip **200** and the second semiconductor chip **300** adjacent (nearest) to the first semiconductor chip **200** may be spaced apart from each other in the third direction **D3**. The second semiconductor chips **300** may be spaced apart from one another in the third direction **D3**. Although the number of the second semiconductor chips **300** stacked over the first semiconductor chip **200** is shown as being two in FIG. 1, the number of the second semiconductor chips **300** is not limited thereto.

(39) Each of the second semiconductor chips **300** may include a third substrate **310**, a third wiring layer **320**, a third lower protective layer **330**, a third upper protective layer **340**, and third through vias **350**.

(40) The third substrate **310** may have the form of a plate extending along a plane defined by the first direction **D1** and the second direction **D2**. In some embodiments, the third substrate **310** may be a semiconductor substrate. For example, the third substrate **310** may include at least one of an elemental semiconductor (e.g., silicon, germanium) and/or a compound semiconductor (e.g., silicon-germanium, and/or a III-V semiconductor (e.g., GaP, and/or GaAs)). In some embodiments, the third substrate **310** may be a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate.

- (41) The third substrate **310** may be provided on the third wiring layer **320**. The third wiring layer **320** may be provided under the third substrate **310**. The third wiring layer **320** may include wirings, and an insulating layer surrounding the wirings. The wirings of the third wiring layer **320** may include a conductive material. In some embodiments, the first wiring layer **120** and the second wiring layer **220** may include the same and/or different materials.
- (42) In some embodiments, the second semiconductor chip **300** may include a memory device. In some embodiments, the second semiconductor chip **300** may include a logic circuit.
- (43) The third wiring layer **320** may be provided on the third lower protective layer **330**. The third lower protective layer **330** may be provided under the third wiring layer **320**. The third lower protective layer **330** may function to protect the third wiring layer **320**. The third lower protective layer **330** may cover a bottom surface of the third wiring layer **320**. The third lower protective layer **330** may include an insulating material. The first lower protective layer **130** and the third lower protective layer **330** may include the same and/or a different insulating material. For example, the third lower protective layer **330** may include silicon nitride and/or silicon oxide.
- (44) The third upper protective layer **340** may be provided on the third substrate **310**. The third upper protective layer **340** may function to protect the third substrate **310**. The third upper protective layer **340** may cover a top surface of the third substrate **310**. The third upper protective layer **340** may include an insulating material. In some embodiments, the first upper protective layer **140** and the third upper protective layer **340** may include the same and/or a different insulating material. For example, the third upper protective layer **340** may include silicon nitride and/or silicon oxide.
- (45) The third through vias **350** may extend through the third substrate **310** and the third upper protective layer **340**. The third through vias **350** may be electrically connected to the wirings of the third wiring layer **320**. The third through vias **350** may include a conductive material.
- (46) A third semiconductor chip **400** may be provided over the plurality of second semiconductor chips **300**. For example, the third semiconductor chip **400** may be provided over the second semiconductor chips **300** in a vertical direction. The third semiconductor chip **400** may be spaced apart from the second semiconductor chip **300** adjacent to the third semiconductor chip **400** in the third direction D3. The first to third semiconductor chips **200**, **300** and **400** may be sequentially stacked over the base structure **100** in the third direction D3. The base structure **100** and the first to third semiconductor chips **200**, **300**, and **400** may overlap in the third direction D3.
- (47) The third semiconductor chip **400** may include a second base structure **410**, a fourth wiring layer **420**, and a fourth lower protective layer **430**.
- (48) The second base structure **410** may have the form of a plate extending along a plane defined by the first direction D1 and the second direction D2. In some embodiments, the second base structure **410** may be a semiconductor substrate. For example, the second base structure **410** may include at least one of an elemental semiconductor (e.g., silicon, germanium) and/or a compound semiconductor (e.g., silicon-germanium, and/or a III-V semiconductor (e.g., GaP, and/or GaAs)). In some embodiments, the second base structure **410** may be a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate.
- (49) The second base structure **410** may be provided on the fourth wiring layer **420**. The fourth wiring layer **420** may be provided under the second base structure **410**. The fourth wiring layer **420** may include wirings, and an insulating layer surrounding the wirings. The wirings of the fourth wiring layer **420** may include a conductive material. In some embodiments, the first wiring layer **120** and the fourth wiring layer **420** may include the same and/or different materials.
- (50) In some embodiments, the third semiconductor chip **400** may include a memory device. In some embodiments, the third semiconductor chip **400** may include a logic circuit.
- (51) The fourth wiring layer **420** may be provided on the fourth lower protective layer **430**. The fourth lower protective layer **430** may be provided under the fourth wiring layer **420**. The fourth lower protective layer **430** may function to protect the fourth wiring layer **420**. The fourth lower

protective layer **430** may cover a bottom surface of the fourth wiring layer **420**. The fourth lower protective layer **430** may include an insulating material. The first lower protective layer **130** and the fourth lower protective layer **430** may include the same and/or a different insulating material. For example, the fourth lower protective layer **430** may include silicon nitride and/or silicon oxide.

(52) Second connecting structures CS2 may be provided between a second semiconductor chip **200** and a neighboring semiconductor chip (e.g., between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, between adjacent ones of the semiconductor chips **200**, and/or between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, respectively).

(53) Each of the second connecting structures CS2 may electrically interconnect the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, may electrically interconnect the adjacent semiconductor chips **200**, and/or may electrically interconnect the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**. The second connecting structures CS2 may include a conductive material. Each of the second connecting structures CS2 may include an upper pad UPD, a lower pad LPD, and/or a connecting bump CBP.

(54) The upper pads UPD of the second connecting structures CS2 may be provided on and/or extend through the third lower protective layers **330** of corresponding ones of the second semiconductor chips **300** or may extend through the fourth lower protective layer **430** of the third semiconductor chip **400**. The upper pads UPD of the second connecting structures CS2 may be electrically connected to the wirings of the third wiring layers **320** of the corresponding second semiconductor chips **300** and/or may be electrically connected to the wirings of the fourth wiring layer **420** of the third semiconductor chip **400**. The upper pads UPD of the second connecting structures CS2 may include a conductive material.

(55) The lower pads LPD of the second connecting structures CS2 may be provided on and/or extend through the second upper protective layer **240** of the first semiconductor chip **200** or may be provided on the third upper protective layers **340** of the corresponding second semiconductor chips **300**. The lower pads LPD of the second connecting structures CS2 may be electrically connected to the second through vias **250** of the first semiconductor chip **200** or may be electrically connected to the third through vias **350** of the corresponding semiconductor chips **300**. The lower pads LPD of the second connecting structures CS2 may include a conductive material.

(56) The connecting bump CBP of each second connecting structure CS2 may be provided between the upper pad UPD and the lower pad LPD in the second connecting structure CS2. The connecting bump CBP of the second connecting structure CS2 may electrically interconnect the upper pad UPD and the lower pad LPD in the second connecting structure CS2. The connecting bump CBP of the second connecting structure CS2 may include a conductive material. For example, the connecting bumps CBP may include a solder ball and/or pillar, and may include a solder, though the example embodiments of the disclosure are not limited thereto.

(57) The first and second connecting structures CS1 and CS2 may electrically interconnect the base structure **100** and the first to third semiconductor chips **200**, **300** and **400**.

(58) Adhesive layers AD may be provided among the first to third semiconductor chips **200**, **300**, and **400**. In some embodiments, the adhesive layers AD may be provided between a second semiconductor chip **300** and an adjacent semiconductor chip (e.g., between the first semiconductor chip **200** and an adjacent second semiconductor chip **200**, between adjacent semiconductor chips **300**, and/or and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, respectively). The first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** may be spaced apart from each other in the third direction D3 with a corresponding one of the adhesive layers AD is interposed therebetween. The adjacent second semiconductor chips **300** may be spaced apart from each other in the third direction D3 with a corresponding one of the adhesive layers AD is

interposed therebetween. The third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400** may be spaced apart from each other in the third direction **D3** with a corresponding one of the adhesive layers **AD** is interposed therebetween. The adhesive layers **AD** may bond the first to third semiconductor chips **200**, **300**, and **400** to each other. The adhesive layers **AD** may surround corresponding ones of the second connecting structures **CS2**, respectively.

(59) A molding layer **MD** covering the first semiconductor chip **200**, the second semiconductor chips **300** and the adhesive layers **AD** may be provided. The molding layer **MD** may be provided on the base structure **100**. The molding layer **MD** may include a base portion surrounding the first to third semiconductor chips **200**, **300**, and **400**, and the adhesive layers **AD**, and an interposition portion **IN** interposed between the base structure **100** and the first semiconductor chip **200**. The base portion **BA** of the molding layer **MD** may surround the interposition portion **IN** of the molding layer **MD**. Although the molding layer **MD** is described under the condition that the molding layer **MD** is divided into the base portion **BA** and the interposition portion **IN** (for convenience of description), the molding layer **MD** may have an integrated structure in which the base portion **BA** and the interposition portion **IN** are connected to each other without having a boundary. The base portion **BA** and the interposition portion **IN** of the molding layer **MD** may include the same material.

(60) The interposition portion **IN** of the molding layer **MD** may surround the first connecting structures **CS1**. The interposition portion **IN** of the molding layer **MD** may be disposed at the same level as the first connecting structures **CS1**. The base structure **100** and the first semiconductor chip **200** may be spaced apart from each other in the third direction **D3** with the interposition portion **IN** of the molding layer **MD** is interposed therebetween. The interposition portion **IN** of the molding layer **MD** may overlap with the base structure **100**, the first to third semiconductor chips **200**, **300**, and **400**, and the adhesive layers **AD** in the third direction **D3**. The width of the interposition portion **IN** of the molding layer **MD** in the first direction **D1** may be equal to and/or defined by the width of the first to third semiconductor chips **200**, **300**, and **400** in the first direction **D1**. For example, the horizontal width of the interposition portion **IN** of the molding layer **MD** may be equal to the horizontal width of the first to third semiconductor chips **200**, **300**, and **400**.

(61) Referring to FIG. **1B**, a bottom surface **1NB** of the interposition portion **IN** of the molding layer **MD** may contact a top surface **TS1** of the base structure **100**. The bottom surface **1NB** of the interposition portion **IN** of the molding layer **MD** may contact a top surface of the first upper protective layer **140** of the base structure **100** (e.g., the top surface **TS1**). A top surface **INT** of the interposition portion **IN** of the molding layer **MD** may contact a bottom surface **BS1** of the first semiconductor chip **200**. The interposition portion **IN** of the molding layer **MD** may cover an overall portion of the bottom surface **BS1** of the first semiconductor chip **200**. The top surface **INT** of the interposition portion **IN** of the molding layer **MD** may contact a bottom surface of the second lower protective layer **230** of the first semiconductor chip **200**, that is, the bottom surface **BS1**. The interposition portion **IN** of the molding layer **MD** may cover an overall portion of the lower surface **BS1** of the second lower protective layer **230**.

(62) The interposition portion **IN** of the molding layer **MD** may contact each first connecting structure **CS1**. A sidewall **INS** of the interposition portion **IN** of the molding layer **MD** may contact a sidewall **SW1** of the first connecting structure **CS1**. The interposition portion **IN** of the molding layer **MD** and the second lower protective layer **230** of the first semiconductor chip **200** may cover an overall portion and/or an exposed portion of the sidewall **SW1** of the first connecting structure **CS1**. The interposition portion **IN** of the molding layer **MD** may contact the upper pad **UPD**, the lower pad **LPD**, and the connecting bump **CBP** of the first connecting structure **CS1**.

(63) The base portion **BA** of the molding layer **MD** may contact a sidewall **SW2** of the first semiconductor chip **200**. The base portion **BA** of the molding layer **MD** may cover an overall portion of a sidewall **SW3** of the second lower protective layer **230** in the first semiconductor chip

200.

(64) The level of the bottom surface **1NB** of the interposition portion **IN** in the molding layer **MD** may be equal to the level of a bottom surface **BS2** of the first connecting structure **CS1**. For example, the bottom surface **1NB** of the interposition portion **IN** of the molding layer **MD** may be coplanar with the bottom surface **BS2** of the first connecting structure **CS1**. The level of the bottom surface **1NB** of the interposition portion **IN** in the molding layer **MD** may be equal to the level of a bottom surface of the lower pad **LPD**, that is, the bottom surface **BS2**, in the first connecting structure **CS1**. The bottom surface **1NB** of the interposition portion **IN** of the molding layer **MD** may be coplanar with the bottom surface **BS2** of the lower pad **LPD** of the first connecting structure **CS1**.

(65) The level of the top surface **INT** of the interposition portion **IN** of the molding layer **MD** may be lower than the level of the top surface **TS2** of the first connecting structure **CS1**. The level of the top surface **INT** of the interposition portion **IN** in the molding layer **MD** may be a level between the top surface **TS2** of the upper pad **UPD** of the first connecting structure **CS1** and a bottom surface of the upper pad **UPD**.

(66) The adhesive layer **AD** interposed between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** from among the adhesive layers **AD** may be defined as a first adhesive layer **AD1**. The first adhesive layer **AD1** may contact a top surface **TS3** of the first semiconductor chip **200** and at least a portion of the sidewall **SW2** of the first semiconductor chip **200**. The first adhesive layer **AD1** may cover an overall portion of the top surface **TS3** of the first semiconductor chip **200**. The first adhesive layer **AD1** may cover an overall portion of a top surface of the second upper protective layer **240**, that is, the top surface **TS3**, in the first semiconductor chip **200**. The first adhesive layer **AD1** may cover an overall portion of a sidewall **SW4** of the second upper protective layer **240** in the first semiconductor chip **200**. The second upper protective layer **240** of the first semiconductor chip **200** may be spaced apart from the base portion **BA** of the molding layer **MD** by the first adhesive layer **AD1**.

(67) The first adhesive layer **AD1** may contact a bottom surface **BS3** and at least a portion of a sidewall **SW5** of the second semiconductor chip **300** adjacent to the first semiconductor chip **200**. The first adhesive layer **AD1** may cover an overall portion of a bottom surface of the third lower protective layer **330**, that is, the bottom surface **BS3**, in the second semiconductor chip **300** adjacent to the first semiconductor chip **200**. The first adhesive layer **AD1** may cover an overall portion of a side wall **SW6** of the third lower protective layer **330** in the second semiconductor chip **300** adjacent to the first semiconductor chip **200**. The third lower protective layer **330** of the second semiconductor chip **300** adjacent to the first semiconductor chip **200** may be spaced apart from the base portion **BA** of the molding layer **MD** by the first adhesive layer **AD1**.

(68) The first adhesive layer **AD1** may contact corresponding ones of the second connecting structures **CS2**. The first adhesive layer **AD** may contact a sidewall of each of the corresponding second connecting structures **CS2**. The level of a lowermost portion of the adhesive layer **AD1** may be lower than the level of the top surface **TS3** of the first semiconductor chip **200**. The level of an uppermost portion of the adhesive layer **AD1** may be higher than the level of the bottom surface **BS3** of the second semiconductor chip **300**.

(69) The adhesive layers **AD** provided between the adjacent second semiconductor chips **300** and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400** may be similar to the first adhesive layer **AD1**.

(70) The distance between the base structure **100** and the first semiconductor chip **200** in the third direction **D3** may be defined as a first distance **L1**. For example, the distance between the top surface **TS1** of the base structure **100** and the bottom surface **BS1** of the first semiconductor chip **200** in the third direction **D3** may be defined as the first distance **L1**. The distance between the bottom surface **1NB** and the top surface **INT** of the interposition portion **IN** of the molding layer **MD** in the third direction **D3** may be equal to the first distance **L1**. The width of the interposition

portion IN of the molding layer MD in the third direction D3 may be equal to the first distance L1. For example, the vertical width of the interposition portion IN of the molding layer MD may be equal to the first distance L1.

(71) The distance between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** in the third direction D3 may be defined as a second distance L2. For example, the distance between the top surface TS3 of the first semiconductor chip **200** and the bottom surface BS3 of the second semiconductor chip **300** adjacent to the first semiconductor chip **200** in the third direction D3 may be defined as the second distance L2. The minimum width of the first adhesive layer AD1 in the third direction D3 may be equal to the second distance L2. For example, the minimum vertical width of the first adhesive layer AD1 may be equal to the second distance L2. The distance between the adjacent second semiconductor chips **300** in the third direction D3 and the distance between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400** in the third direction D3 may be equal to the second distance L2.

(72) The first distance L1 and the second distance L2 may be different. In some embodiments, the first distance L1 may be smaller than the second distance L2. In some embodiments, the first distance L1 and the second distance L2 may be about 5 to about 20 μm .

(73) Referring to FIG. 1C, the molding layer MD may include a first material layer ML1 and first fillers FI1. Each of the base portion BA and the interposition portion IN of the molding layer MD may include a part of the first material layer ML1 and the first fillers FI1. The first material layer ML1 may surround the first fillers FI1. For example, the first fillers FI1 may be provided in the first material layer ML1. The first material layer ML1 may include a polymer material. For example, the first material layer ML1 may include an epoxy resin. The first fillers FI1 may include a ceramic material. For example, the first fillers FI1 may include silicon oxide.

(74) In some example embodiments, the content of the first fillers FI1 of the molding layer MD may be 70 wt % or more. In some example embodiments, the molding layer MD may have a Young's modulus of about 5 to about 20 GPa at 25° C. For example, the molding layer MD may have a glass transition temperature of 90° C. to 200° C. For example, the molding layer MD may have a coefficient of thermal expansion of 5 to 40 ppm at a temperature not higher than the glass transition temperature and may have a coefficient of thermal expansion of 20 to 100 ppm at a temperature higher than the glass transition temperature. In some example embodiments, the maximum diameter of the first fillers FI1 may be 5 μm or less.

(75) At least one of the adhesive layers (e.g., the first adhesive layer AD1) may include a material different from that of the molding layer MD. A boundary dividing the first adhesive layer AD1 and the molding layer MD from each other may be formed between the first adhesive layer AD1 and the molding layer MD. The first adhesive layer AD1 may include a second material layer ML2 and second fillers FI2. The second material layer ML2 may surround the second fillers FI2. For example, the second fillers FI2 may be provided in the second material layer ML2. The second material layer ML2 may include a material different from that of the first material layer ML1. The second material layer ML2 may include an adhesive polymer material. The second fillers FI2 may include a ceramic material. For example, the second fillers FI2 may include silicon oxide.

(76) In some example embodiments, the content of the first fillers FI1 of the molding layer MD may be 1.5 to 2.5 times the content of the second fillers FI2 of the first adhesive layer AD1. In some example embodiments, the average diameter of the second fillers FI2 may be smaller than the average diameter of the first fillers FI1.

(77) The adhesive layers AD provided between the adjacent semiconductor chips **300** and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400** may include the second material layer ML2 and the second fillers FI2, substantially identically to the first adhesive layer AD1.

(78) In the semiconductor package according to some example embodiments of the disclosure, a

triple point where the protective layer (e.g., the upper protective layer and/or the lower protective layer), the molding layer MD and the adhesive layer AD contact may not be formed because the interposition portion IN of the molding layer MD is provided between the base structure **100** and the first semiconductor chip **200**. Accordingly, a phenomenon in which stress is concentrated on the triple point may be prevented and, as such, reliability of the semiconductor package may be enhanced.

(79) FIGS. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, 2K, and 2L are sectional views explaining a method for manufacturing a semiconductor package in accordance with some example embodiments of the disclosure.

(80) Referring to FIG. 2A, a base structure **100** including a first substrate **110**, a first wiring layer **120**, a first lower protective layer **130**, and first through vias **150** may be formed. In some embodiments, formation of the first structure **110** may include forming the first substrate **110**, forming the first through vias **150** in the first substrate **110**, forming the first wiring layer **120** on the first substrate **110**, and forming the first lower protective layer **130** on the first wiring layer **120**. Base pads BPD may be formed on the base structure **100**. Base bumps BBP may be formed on the base pads BPD, respectively.

(81) Referring to FIG. 2B, the base structure **100** may be attached to a first carrier substrate CAR1 using a first glue layer GL1.

(82) The first lower protective layer **130** of the base structure **100** may contact the first glue layer GL1. The base bumps BBP may be surrounded by the first glue layer GL1. For example, the first glue layer GL1 may include an acrylic polymer.

(83) Referring to FIG. 2C, a portion of the first substrate **110** of the base structure **100** may be removed. For example, the portion of the first substrate **110** may be removed through an etch-back process and/or a grinding process. As the portion of the first substrate **110** is removed, an end of the first through vias **150** may be exposed. A first upper protective layer **140** may be formed on the first substrate **110**. In some embodiments, formation of the first upper protective layer **140** may include forming, on the first substrate **110**, a first preliminary upper protective layer covering the first through vias **150**, and removing an upper portion of the first preliminary upper protective layer, thereby forming the first upper protective layer **140** exposing the first through vias **150**. Lower pads LPD may be formed on the first through vias **150**, respectively.

(84) Referring to FIG. 2D, a first preliminary semiconductor chip p**200** including a second substrate **210**, a second wiring layer **220**, a second lower protective layer **230**, second through vias **250**, upper pads UPD, and connecting bumps CBP may be formed through processes similar to the above-described processes. The first preliminary semiconductor chip p**200** may be attached to a second carrier substrate CAR2 using a second glue layer GL2. A second upper protective layer **240** may be formed on the second substrate **210**. Lower pads LPD may be formed on the second through vias **250**, respectively.

(85) A first tape TP1 covering the second upper protective layer **240** of the first preliminary semiconductor chip p**200** and the lower pads LPD may be formed. The first tape TP1 may contact the second upper protective layer **240** of the first preliminary semiconductor chip p**200** and the lower pads LPD.

(86) Referring to FIG. 2E, the first preliminary semiconductor chip p**200** may be inverted using the first tape TP1. As the first preliminary semiconductor chip p**200** is inverted, the second carrier substrate CAR2 and the second glue layer GL2 may be disposed over the first preliminary semiconductor chip p**200**.

(87) The second carrier substrate CAR2 and the second glue layer GL2 may be removed. As the second carrier substrate CAR2 and the second glue layer GL2 are removed, the upper pads UPD and the connecting bumps CBP on the first preliminary semiconductor chip p**200** may be exposed.

(88) Referring to FIG. 2F, a second tape TP2 covering the second lower protective layer **230** of the first preliminary semiconductor chip p**200**, the upper pads UPD and the connecting bumps CBP

may be formed. The second tape TP2 may contact the second lower protective layer 230 of the first preliminary semiconductor chip p200, the upper pads UPD and the connecting bumps CBP. The first preliminary semiconductor chip p200 may be inverted using the first tape TP1 and the second tape TP2. As the first preliminary semiconductor chip p200 is inverted, the first tape TP1 may be disposed over the first preliminary semiconductor chip p200.

(89) In some embodiments, it may be possible to remove the second glue layer GL2 and the second carrier substrate CAR2 without using the process of inverting the first preliminary semiconductor chip p200, differently from the above-described procedure, and, as such, the second tape TP2 may be formed.

(90) Referring to FIG. 2G, the first preliminary semiconductor chip p200 may be divided into a plurality of first semiconductor chips 200. In some embodiments, division of the first preliminary semiconductor chip p200 into the plurality of first semiconductor chips 200 may include removing the first tape TP1 on the first preliminary semiconductor chip p200, and dividing the first preliminary semiconductor chip p200 into the plurality of first semiconductor chips 200. The first preliminary semiconductor chip p200 may be divided into the first semiconductor chips 200 through a dicing process. The divided first semiconductor chips 200 may be connected by the second tape TP2.

(91) Division of the first preliminary semiconductor chip p200 into the plurality of first semiconductor chips 200 may include dividing the second upper protective layer 240 into a plurality of pieces, dividing the second substrate 210 into a plurality of pieces, dividing the second wiring layer 220 into a plurality of pieces, and dividing the second lower protective layer 230 into a plurality of pieces.

(92) Referring to FIG. 2H, each of the first semiconductor chips 200 may be provided over the base structure 100. Each first semiconductor chip 200 may be mounted on the base structure 100.

Provision of the first semiconductor chip 200 over the base structure 100 may include separating the first semiconductor chip 200 from the second tape TP2, and bonding the connecting bumps CBP connected to the first semiconductor chip 200 and lower pads LPD connected to the base structure 100, respectively. For example, the connecting bumps CBP connected to the first semiconductor chip 200 and the lower pads LPD connected to the base structure 100 may be bonded to each other through a thermal compression process, a mass reflow process and/or a laser assisted bonding process.

(93) As the connecting bumps CBP connected to the first semiconductor chip 200 and the lower pads LPD connected to the base structure 100 are bonded to each other, first connecting structures CS1 each including corresponding ones of the upper pads UPD, the connecting bumps CBP, and the lower pads LPD may be formed.

(94) The first semiconductor chip 200 and the base structure 100 may be spaced apart from each other in a third direction D3 under the condition that the first connecting structures CS1 are interposed therebetween. An empty space provided between the first semiconductor chip 200 and the base structure 100 may be defined as a cavity CA. A space between a bottom surface of the first semiconductor chip 200 and a top surface of the base structure 100 may be defined as the cavity CA.

(95) The first connecting structures CS1 may be exposed by the cavity CA. The top surface of the base structure 100 and the bottom surface of the first semiconductor chip 200 may be exposed by the cavity CA. The cavity CA may expose an overall portion of the bottom surface of the first semiconductor chip 200. The cavity CA may expose an overall portion of a bottom surface of the second lower protective layer 230.

(96) Referring to FIG. 2I, a second preliminary semiconductor chip p300 including a third substrate 310, a third wiring layer 320, a third lower protective layer 330, a third upper protective layer 340, third through vias 350, upper pads UPD, lower pads LPD, and connecting bumps CBP may be formed through processes similar to the above-described processes. A third tape TP3 covering the

third upper protective layers **340** of the second preliminary semiconductor chip **p300** and the lower pads **LPD** may be formed. The third tape **TP3** may contact the third upper protective layer **340** of the second preliminary semiconductor chip **p300** and the lower pads **LPD**.

(97) A preliminary adhesive layer **pAD** covering the third lower protective layer **330** of the second preliminary semiconductor chip **p300**, the upper pads **UPD** and the connecting bumps **CBP**, and a fourth tape **TP4** supporting the preliminary adhesive layer **pAD** may be formed. The preliminary adhesive layer **pAD** may contact the third lower protective layer **330** of the second preliminary semiconductor chip **p300**, the upper pads **UPD**, and the connecting bumps **CBP**. The fourth tape **TP4** may be spaced apart from the third lower protective layer **330** of the second preliminary semiconductor chip **p300**, the upper pads **UPD** and the connecting bumps **CBP**. In some embodiments, formation of the fourth tape **TP4** and the preliminary adhesive layer **pAD** may include providing the fourth tape **TP4**, forming the preliminary adhesive layer **pAD** on the fourth tape **TP4**, and attaching the second preliminary semiconductor chip **p300** to the preliminary adhesive layer **pAD**. For example, the preliminary adhesive layer **pAD** may include an adhesive polymer material and a ceramic material.

(98) Referring to FIG. 2J, the second preliminary semiconductor chip **p300** may be divided into a plurality of second semiconductor chips **300**, and the preliminary adhesive layer **pAD** may be divided into a plurality of adhesive layers **AD**. For example, the second preliminary semiconductor chip **p300** and the preliminary adhesive layer **pAD** may be simultaneously divided into the second semiconductor chips **300** and the adhesive layers **AD** through one-time execution of a dicing process. In some embodiments, the second preliminary semiconductor chip **p300** and the preliminary adhesive layer **pAD** may be divided after removal of the third tape **TP3**. The divided second semiconductor chips **300** and the divided adhesive layers **AD** may be connected by the fourth tape **TP4**.

(99) Referring to FIG. 2K, the second semiconductor chips **300** may be provided over the first semiconductor chips **200**, respectively. The second semiconductor chips **300** may be mounted on the first semiconductor chips **200**, respectively. Provision of the second semiconductor chips **300** over the first semiconductor chips **200** may include separating the second semiconductor chips **300** and the adhesive layers **AD** from the fourth tape **TP4**, bonding the connecting bumps **CBP** connected to each of the second semiconductor chips **300** and the lower pads **LPD** connected to a corresponding one of the first semiconductor chips **200**, attaching the second semiconductor chip **300** and the first semiconductor chip **200** using a corresponding one of the adhesive layers **AD**. For example, the connecting bumps **CBP** connected to the second semiconductor chip **300** and the lower pads **LPD** connected to the first semiconductor chip **200** may be bonded to each other through a thermal compression process and/or a mass reflow process.

(100) As the second semiconductor chip **300** is provided over the first semiconductor chip **200**, the shape of the adhesive layer **AD** between the second semiconductor chip **300** and the first semiconductor chip **200** may be varied. As the connecting bumps **CBP** connected to the second semiconductor chip **300** and the lower pads **LPD** connected to the first semiconductor chip **200** are bonded to each other, second connecting structures **CS2** each including corresponding ones of the upper pads **UPD**, the connecting bumps **CBP** and the lower pads **LPD** may be formed.

(101) Referring to FIG. 2L, the second semiconductor chips **300** and a third semiconductor chip **400** may be sequentially provided over the base structure **100** through processes similar to the above-described processes. Adhesive layers **AD** may be formed between adjacent ones of the second semiconductor chips and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, respectively. The base structure **100**, the first to third semiconductor chips **200**, **300** and **400**, and the adhesive layers **AD** may overlap with the cavity **CA** (cf. FIG. 2H) in the third direction **D3**.

(102) A molding layer **MD** covering the top surface of the base structure **100** and the first to third semiconductor chips **200**, **300**, and **400** may be formed. The molding layer **MD** may fill the cavity

CA between the first semiconductor chip **200** and the base structure **100**. An interposition portion IN of the molding layer MD may fill the cavity CA between the first semiconductor chip **200** and the base structure **100**.

(103) The first carrier substrate CAR1 and the first glue layer GL1 may be separated from the base structure **100**. After separation of the first carrier substrate CAR1 and the first glue layer GL1 from the base structure **100**, the base structure **100** may be divided into a plurality of base structures **100**, and the molding layer MD may be divided into a plurality of molding layers MD. For example, the base structure **100** and the molding layer MD may be divided through a dicing process.

(104) In the semiconductor package manufacturing method according to the example embodiments of the disclosure, the adhesive layers AD are formed between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, between the adjacent second semiconductor chips **300**, and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, respectively, and the interposition portion IN of the molding layer MD may be formed between the first semiconductor chip **200** and the base structure **100**. Since the number of regions where the interposition portion IN of the molding layer MD is formed is relatively small, difficulty of the process for forming the molding layer MD may be relatively small.

(105) FIG. 3 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(106) Referring to FIG. 3, a semiconductor package 2 may include a molding layer MDa including a base portion Baa and an interposition portion INa. The interposition portion INa of the molding layer MDa may be interposed between a third semiconductor chip **400** and a second semiconductor chip **300** adjacent to the third semiconductor chip **400**. The semiconductor package 2 may include adhesive layers ADa. The adhesive layers ADa may be interposed between a base structure **100** and a first semiconductor chip **200**, between the first semiconductor chip **200** and a second semiconductor chip **300** adjacent to the first semiconductor chip **200**, and between the second semiconductor chips **300**, respectively.

(107) FIG. 4 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(108) Referring to FIG. 4, a semiconductor package 3 may include a molding layer MDb including a base portion Bab and an interposition portion INb. The interposition portion INb of the molding layer MDb may be interposed between second semiconductor chips **300**. The semiconductor package 3 may include adhesive layers ADb. The adhesive layers ADb may be interposed between a base structure **100** and a first semiconductor chip **200**, between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, and between a third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, respectively.

(109) Although the interposition portion INb of the molding layer MDb is shown and described as being interposed between the second semiconductor chips **300**, the example embodiments of the disclosure are not limited thereto. In some embodiments, the interposition portion INb of the molding layer MDb may be provided between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, differently from the case as shown in FIG. 4.

(110) FIG. 5 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(111) Referring to FIG. 5, a semiconductor package 4 may include a first adhesive layer ADc1 and second adhesive layers ADc2. The first adhesive layer ADc1 may be interposed between a first semiconductor chip **200** and a second semiconductor chip **300** adjacent to the first semiconductor chip **200** from among second semiconductor chips **300**. The second adhesive layers ADc2 may be interposed between adjacent ones of the second semiconductor chips **300** and between a third

semiconductor chip **400** and a second semiconductor chip **300** adjacent to the third semiconductor chip **400** from among the second semiconductor chips **300**, respectively.

(112) The first adhesive layer ADc1 may include a material different from that of each second adhesive layer ADc2. For example, the first adhesive layer ADc1 and the second adhesive layer ADc2 may include different adhesive polymer materials and/or different concentrations of ceramics, respectively. The distance between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** in a third direction D3 may differ from the distance between the adjacent second semiconductor chips **300** in the third direction D3. The distance between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** in the third direction D3 may differ from the distance between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400** in the third direction D3.

(113) The minimum width of the first adhesive layer ADc1 in the third direction D3 may differ from the minimum width of the second adhesive layer ADc2 in the third direction D3. For example, the minimum vertical width of the first adhesive layer ADc1 may differ from the minimum vertical width of the second adhesive layer ADc2.

(114) Although the first adhesive layer ADc1 is shown and described as being interposed between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200**, and the second adhesive layers ADc2 are being shown and described as being interposed between the adjacent second semiconductor chips **300** and between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**, the example embodiments of the disclosure are not limited thereto. For example, in some embodiments, first adhesive layers ADc1 may be interposed between the first semiconductor chip **200** and the second semiconductor chip **300** adjacent to the first semiconductor chip **200** and between the adjacent second semiconductor chips **300**, respectively, and a second adhesive layer ADc2 may be interposed between the third semiconductor chip **400** and the second semiconductor chip **300** adjacent to the third semiconductor chip **400**.

(115) FIG. 6 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(116) Referring to FIG. 6, a semiconductor package 5 may include a base structure **100d**, and a first semiconductor chip **200d**, second semiconductor chips **300d** and a third semiconductor chip **400d** that are sequentially stacked over the base structure **100d**. Adhesive layers Add may be interposed between the first semiconductor chip **200d** and the second semiconductor chip **300d** adjacent to the first semiconductor chip **200d**, between adjacent ones of the second semiconductor chips **300d** and between the third semiconductor chip **400d** and the second semiconductor chip **300d** adjacent to the third semiconductor chip **400d**, respectively. Although the number of the first to third semiconductor chips **200d**, **300d**, and **400d** is shown and described as being eight, the example embodiments of the disclosure are not limited thereto.

(117) FIG. 7 is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(118) Referring to FIG. 7, a semiconductor package 6 may include a base structure **100e**, and a first semiconductor chip **200e**, second semiconductor chips **300e** and a third semiconductor chip **400e** which are sequentially stacked over the base structure **100e**.

(119) A molding layer MDe may include a base portion Bae, and a plurality of interposition portions INe connected to the base portion Bae. Adhesive layers Ade and the interposition portions INe may be randomly interposed in spaces among the base structure **100e** and the first to third semiconductor chips **200e**, **300e**, and **400e**. For example, as shown in FIG. 7, five adhesive layers Ade and three interposition portions INe may be interposed in the spaces among the base structure **100e** and the first to third semiconductor chips **200e**, **300e** and **400e**, respectively. However, the positions and the numbers of the adhesive layers Ade and the interposition portions INe interposed

in the spaces among the base structure **100e** and the first to third semiconductor chips **200e**, **300e** and **400e**.

(120) FIG. **8** is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(121) Referring to FIG. **8**, a semiconductor package **7** may include a package substrate **500**. For example, the package substrate **500** may be a printed circuit board (PCB). External terminals **510** electrically connected to the package substrate **500** may be provided. The external terminals **510** may be provided under the package substrate **500**. The external terminals **510** may include a conductive material. The semiconductor package **7** may be mounted on an external device (for example, a main board) through the external terminals **510**.

(122) An interposer **600** may be provided over the package substrate **500**. First connecting terminals **610** electrically interconnecting the package substrate **500** and the interposer **600** may be provided. The first connecting terminals **610** may include a conductive material. A first underfill layer **620** may be provided between the package substrate **500** and the interposer **600**.

(123) A processor chip **700** may be provided over the interposer **600**. The processor chip **700** may include logic circuits. For example, the processor chip **700** more specifically may include, but is not limited to, a graphics processing unit (GPU) central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), and programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc. Second connecting terminals **710** electrically interconnecting the processor chip **700** and the interposer **600** may be provided. The second connecting terminals **710** may be provided between the processor chip **700** and the interposer **600**. The second connecting terminals **710** may include a conductive material. A second underfill layer **720** may be provided between the processor chip **700** and the interposer **600**.

(124) A base structure **100f** and first to third semiconductor chips **200f**, **300f**, and **400f** may be sequentially provided over the interposer **600** in a third direction D3. The base structure **100f** and the first to third semiconductor chips **200f**, **300f**, and **400f** may be spaced apart from the processor chip **700** in a first direction D1. A third underfill layer **820** may be provided between the base structure **100f** and the interposer **600**.

(125) Adhesive layers ADf may be provided between the first semiconductor chip **200f** and the second semiconductor chip **300f** adjacent to the first semiconductor chip **200f**, between adjacent ones of the second semiconductor chips **300f** and between the third semiconductor chip **400f** and the second semiconductor chip **300f** adjacent to the third semiconductor chip **400f**, respectively.

(126) A first molding layer MD1 covering the first semiconductor chip **200f**, the second semiconductor chips **300f** and the adhesive layers ADf may be provided. The first molding layer MD1 may include a base portion BA1 and an interposition portion IN1. The interposition portion IN1 of the first molding layer MD1 may be interposed between the base structure **100f** and the first semiconductor chip **200f**. The first molding layer MD1 may contact a first connecting structure CS1f between the base structure **100f** and the first semiconductor chip **200f**. The interposition portion IN1 of the first molding layer MD1 may contact the first connecting structure CS1f between the base structure **100f** and the first semiconductor chip **200f**. The first molding layer MD1 may include a material different from those of the first to third underfill layers **620**, **720**, and **820**.

(127) A second molding layer MD2 covering the first to third semiconductor chips **200f**, **300f** and **400f** and the processor chip **700** may be provided. The second molding layer MD2 may cover a top surface of the interposer **600**. The second molding layer MD2 may contact the second underfill layer **720** and the third underfill layer **820**. A boundary dividing the second molding layer MD2 and the second underfill layer **720** from each other may be formed between the second molding layer MD2 and the second underfill layer **720**. A boundary dividing the second molding layer MD2 and the third underfill layer **820** from each other may be formed between the second molding layer MD2 and the third underfill layer **820**. The second molding layer MD2 may include a material

different from those of the first to third underfill layers **620**, **720** and **820**.

(128) FIG. **9** is a sectional view of a semiconductor package according to some example embodiments of the disclosure.

(129) Referring to FIG. **9**, a semiconductor package **8** may include a package substrate **500**, external terminals **510**, an interposer **600**, first connecting terminals **610**, a first underfill layer **620**, a processor chip **700**, and second connecting terminals **710**.

(130) A base structure **100g** and first to third semiconductor chips **200g**, **300g**, and **400g** may be sequentially provided over the interposer **600** in a third direction D3. Adhesive layers ADg may be provided between the base structure **100g** and the first semiconductor chip **200g**, between the first semiconductor chip **200g** and the second semiconductor chip **300g** adjacent to the first semiconductor chips **200g**, between adjacent ones of the second semiconductor chips **300g** and between the third semiconductor chip **400g** and the second semiconductor chip **300g** adjacent to the third semiconductor chip **400g**, respectively.

(131) A first molding layer MD1g covering the first semiconductor chip **200g**, the second semiconductor chips **300g** and the adhesive layers ADg may be provided. The first molding layer MD1g may be spaced apart from a first connecting structure CS1g between the base structure **100g** and the first semiconductor chip **200g**. A portion of the adhesive layer ADg corresponding to the first connecting structure CS1g may be interposed between the first molding layer MD1g and the first connecting structures CS1g.

(132) A second molding layer MD2g covering the base structure **100g**, the first to third semiconductor chips **200g**, **300g** and **400g** and the processor chip **700** may be provided. The second molding layer MD2g may include a base portion BA2, and interposition portions IN2 connected to the base portion BA2. Each interposition portion BA2 may be interposed between the base structure **100g** and the interposer **600** or between the processor chip **700** and the interposer **600**. Each interposition portion IN2 may surround the second connecting terminals **710** or base bumps BBPg.

(133) In the semiconductor package according to some example embodiments of the disclosure, a triple point where the molding layer, the adhesive layer, and the protective layer contact may not be formed because the interposition portion of the molding layer is provided. Accordingly, reliability of the semiconductor package may be enhanced.

(134) While the some example embodiments of the disclosure have been described with reference to the accompanying drawings, it should be understood by those skilled in the art that various modifications may be made without departing from the scope of the disclosure and without changing essential features thereof. Therefore, the above-described embodiments should be considered in a descriptive sense only and not for purposes of limitation.

Claims

1. A method of manufacturing a semiconductor package, comprising: forming a base structure; stacking a first semiconductor chip over the base structure, a cavity being defined between the base structure and the first semiconductor chip; stacking a second semiconductor chip over the first semiconductor chip, using an adhesive layer; and forming a molding layer covering the first semiconductor chip, the second semiconductor chip and the adhesive layer, wherein the forming the molding layer includes filling the cavity with an interposition portion of the molding layer.
2. The method of manufacturing the semiconductor package according to claim 1, wherein the stacking the first semiconductor chip over the base structure comprises forming a connecting structure electrically interconnecting the base structure and the first semiconductor chip, and wherein the connecting structure is exposed by the cavity.
3. The method of manufacturing the semiconductor package according to claim 1, wherein the stacking the second semiconductor chip over the first semiconductor chip comprises providing the

adhesive layer between the first semiconductor chip and the second semiconductor chip.

4. The method of manufacturing the semiconductor package according to claim 1, wherein a bottom surface of the first semiconductor chip includes at least one pad and a portion including a remainder of the bottom surface, and wherein the cavity exposes the portion of the bottom surface of the first semiconductor chip.
 5. The method of manufacturing the semiconductor package according to claim 1, wherein the first semiconductor chip includes a lower protective layer, and wherein the cavity exposes an overall portion of a bottom surface of the lower protective layer.
 6. The method of manufacturing the semiconductor package according to claim 1, wherein the cavity is provided between a bottom surface of the first semiconductor chip and a top surface of the base structure.
 7. The method of manufacturing the semiconductor package according to claim 1, wherein the stacking the first semiconductor chip over the base structure comprises, covering a first preliminary semiconductor chip with a first tape, dividing the first preliminary semiconductor chip into a plurality of first semiconductor chips, and separating the first semiconductor chip from the first tape.
 8. The method of manufacturing the semiconductor package according to claim 1, wherein the stacking the second semiconductor chip over the first semiconductor chip comprises, providing a second preliminary semiconductor chip, providing a preliminary adhesive layer and a second tape, the preliminary adhesive layer covering a the second preliminary semiconductor chip and a the second tape supporting the preliminary adhesive layer, dividing the second preliminary semiconductor chip into a plurality of second semiconductor chips and the preliminary adhesive layer into a plurality of adhesive layers, and separating the second semiconductor chip and the preliminary adhesive layer from the second tape.
 9. The method of manufacturing the semiconductor package according to claim 1, wherein the cavity vertically overlaps with the base structure, the first semiconductor chip and the second semiconductor chip.
 10. A method of manufacturing a semiconductor package, comprising: forming a base structure, comprising a lower pad; forming a first semiconductor chip, comprising a connecting bump and an upper pad; forming a first connecting structure, by interconnecting the lower pad, the connecting bump and the upper pad, stacked sequentially; stacking a second semiconductor chip over the first semiconductor chip, using an adhesive layer; and forming a molding layer covering the first semiconductor chip, the first connecting structure, the second semiconductor chip and the adhesive layer, wherein the molding layer includes an interposition portion, which contacts the first connecting structure.
 11. The method of manufacturing the semiconductor package according to claim 10, wherein the interposition portion contacts a top surface of the base structure and a bottom surface of the first semiconductor chip.
 12. The method of manufacturing the semiconductor package according to claim 10, wherein the base structure is isolated from direct contact with the first semiconductor chip; and a vertical distance between the base structure and the first semiconductor chip is 5 to 20 μm .
 13. The method of manufacturing the semiconductor package according to claim 10, wherein the interposition portion entirely covers a bottom surface of the first semiconductor chip.
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